

FixHeatmap v2 - Functional and Technical Specification

1. Objective

The application supports intraday trading decisions between Fixing1 and Fixing2 on the Polish power market. The objective is to identify hours where historical behaviour indicates a statistically and economically meaningful trading edge. The application shall combine directional probability and spread magnitude into a single Edge Score.

2. Data Source

Input data source: fix1_fix2_delta_H history (hourly). Required fields: - Delivery Date - Hour (1-24) - Fixing1 Price - Fixing2 Price - Spread = Fixing2 - Fixing1 Minimum history: 6 months. Current calibration based on approximately 4,223 hourly observations and 171 delivery days.

3. Existing Heatmap

The current heatmap remains unchanged. Green cell: Spread > 0 (Buy F1 / Sell F2) Red cell: Spread < 0 (Sell F1 / Buy F2) Cell value: Actual spread between Fixing2 and Fixing1.

4. Probability Module

For a user-selected rolling window (3, 5, 7, 14, 30 days): ProbabilityLONG = Number of days with positive spread / Total number of days ProbabilitySHORT = 1 - ProbabilityLONG Display probability chart by hour.

5. Edge Score Calculation

For each hour: Edge = $(2 \times \text{ProbabilityLONG} - 1) \times P75(|\text{Spread}|)$ Where: ProbabilityLONG = historical probability of positive spread $P75(|\text{Spread}|)$ = 75th percentile of absolute spread within the selected window Reason: P75 better reflects economically meaningful opportunities than median spread and is less sensitive to outliers than maximum spread.

6. Edge Interpretation

Edge > 0 -> LONG bias Edge < 0 -> SHORT bias Absolute value determines signal strength.

7. Trading Legend

$|\text{Edge}| < 3$ -> No edge / No trade $3 \leq |\text{Edge}| < 7$ -> Weak signal $7 \leq |\text{Edge}| < 12$ -> Moderate signal $12 \leq |\text{Edge}| < 18$ -> Strong signal $18 \leq |\text{Edge}| < 25$ -> Very strong signal $|\text{Edge}| \geq 25$ -> Exceptional signal Current thresholds calibrated using 6-month market history.

8. New Dashboard Panel

Add a third chart below the existing Probability chart. Chart name: EDGE SCORE Requirements: - X axis = Hour (1-24) - Y axis = Edge Score - Positive values = Green bars - Negative values = Red bars - Show numeric Edge value above each bar - Show tooltip with: Hour ProbabilityLONG P75

Spread Edge Score

9. Trading Ranking

Generate ranking table: Hour Direction (LONG/SHORT) ProbabilityLONG P75 Spread Edge Score
Signal Strength Sort descending by absolute Edge Score.

10. Future Integration with Forecast Models

Architecture must allow future integration of forecasting models (XGBoost, Random Forest, SARIMAX, ARIMA). Future version: Edge Score will be combined with model directional accuracy and model PnL performance. Current version uses historical Fixing1 vs Fixing2 statistics only.